

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Chemistry

2. Subject Name: Analytical and Nuclear Chemistry L-T-P: 3-0-0 Credits: 3

3. Pre-requisites: NIL

4. Syllabus and reference books:

Syllabus:

1. Introductory Topics: Analytical terms, precision and accuracy, significant figures, types of errors and treatment of errors. Validation of experimental methods and standard reference materials (6L).

- Measurements: Signal-to-Noise Ratio, Origin of instrument noise and quantifying the measurements (2L).

2. Thermal Analysis: Basics of instrumentation of thermogravimetric analysis (TGA), differential thermal analysis (DTA) and differential scanning calorimetry (DSC), and their applications in chemistry (4L).

3. Photo Electron Spectroscopy: Fundamentals of photoelectron spectroscopy (PES) – XPS and UPS methods and their applications. Auger electron microscopy (AES) for surface analysis of materials (4L).

4. Chromatographic Separations: Basic principles in chromatography including paper, thin layer and column chromatography. Theory behind chromatographic parameters and resolution (2L).

- Liquid Chromatography: Fundamentals of liquid chromatograph - HPLC instrumentation, chemistry behind stationary phase and eluents, classifications on adsorption, partition and ion-exchange chromatography, etc. Isocratic and gradient methods of separation with suitable examples (3L).

- Gas Chromatography: Basic description of gas chromatography, classification of GC methods, stationary phase, carrier gas, temperature programming and different detectors used in chemical analysis (3L).

6. Mass Spectrometry (MS): Fundamental concepts of modern mass spectrometry and different types of mass spectrometers (like LC-MS, GC-MS, HRMS and ICP-MS), ionization techniques used in MS, isotopic patterns, fragmentation rules and interpretation of mass spectra using suitable examples (8L).

7. Nuclear Chemistry: Fundamental aspects and Applications like radiocarbon dating, neutron activation analysis (NAA) and accelerator mass spectrometry (AMS) (10L).

Reference Books:

(1) Fundamentals of Analytical Chemistry, 9th Edn – D. A. Skoog, Donald M. West, F. J. Holler, S. R. Crouch.

(2) Principles of Instrumental Analysis, 7th Edn. – Douglas A. Skoog, F. James Holler and S. R. Crouch.

(3) Inorganic Chemistry. 2nd Edn, Pearson Education Ltd, -Catherine E. Housecroft and Alan G. Sharpe.

(4) Quantitative Chemical Analysis, Daniel C. Harris, 9th edition, 2015.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introductory Topics	08
2.	Thermal Analysis	04
3.	Photo Electron Spectroscopy	04
4.	Chromatographic Separations	08
5.	Mass Spectrometry (MS)	08
6.	Nuclear Chemistry	10
Total number of hours		42